



Shared Assets

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1. Introduction

Welcome. Here you will find information about the shared assets that are included in the assets published by Nicrom. If you have a question or encountered a problem for which you did not find an answer here, please contact me using one of the support methods listed in the [Social Links](#) chapter.

2. Global Managers

The assets published by Nicrom include several global managers: *Environment Manager*, *Post Processing Volume Manager*, *Wind Manager* and *SGP Manager*. In the asset's *Prefabs -> Managers* folder you will find a prefab called *GlobalManagers*. This prefab serves as a container for all the global managers.

If you use third party vegetation you will not need to have the *Wind Manager* and *SGP Manager* in your scene as they only work with material that use shaders created by Nicrom. These managers can be disabled or removed from *GlobalManagers* prefab. Alternatively, you can create your own *GlobalManagers prefab* that contains only the managers that you need.

2.1. Environment Manager

The *Environment Manager* controls the properties of the *main light*, the *ambient light*, the *environment fog* and *skybox*. A single *Environment Manager* object must exist in a scene. If multiple scenes are used, only one scene must have the *Environment Manager* in it. To add the *Environment Manager* to a scene:

- Create an empty game object and add *Environment Manager* script to it, Figure 1.
- Right click in the *Project* window and in the context menu go to *Create -> Nicrom* and click on *Environment Profile*. An EP is a scriptable object that contains the environment properties that you can customize. If you already have an EP asset, you do not need to create a new one.
- Drag and drop the EP asset in the *Profile* property of the *Environment Manager*.
- Drag and drop the main directional light in the *Main Light* property.
- Drag and drop the moon directional light in the *Moon Light* property. If moon light is not needed, this field can be left empty.
- If you use a *Post Processing Volume Manager*, drag and drop it in the *PPV Manager* property. If not, this field can be left empty.

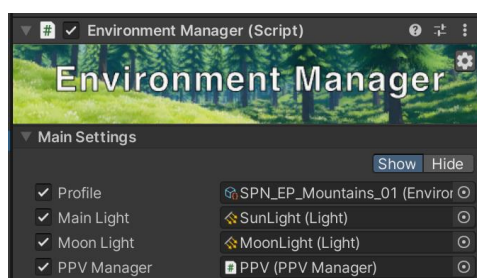


Figure 1

The properties of an **Environment Profile** are divided in 3 tabs: *Light*, *Fog* and *Skybox*.

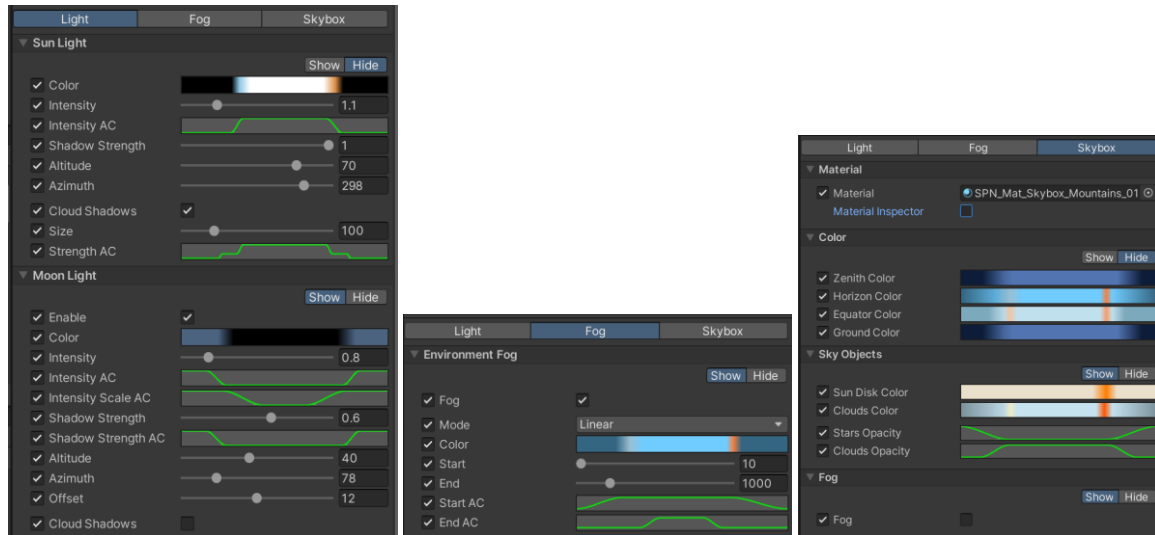


Figure 2

To the left of a property you will see an **override** toggle. When the override is disabled, the property will become locked and the user will not be able to change its value.

Some of the environment settings can have a float property and an animation curve property in the EP. For example if we look at the *Fog* tab, Figure 2, the fog *End* has a float slider and an animation curve. In this case when the EM will multiply the value of the float property with that of the AC. If one is disabled it will use only the enabled, one.

The **Show/Hide** buttons can be used to show/hide in the inspector the properties that are disabled.

Sun/Moon Light – when the EM detects that the sun intensity is 0, it will disable the light component of the sun light and enable the moon light component if the moon light feature is enabled in the EP settings.

2.2. Wind Manager

The *Wind Manager* controls the global properties of the wind simulation. If there is no *Wind Manager* in a scene, the wind will not be simulated. A single *Wind Manager* must exist in a scene. If multiple scenes are used, only one scene must have the *Wind Manager* in it. To add a WM to a scene:

- Create an empty game object and add *Wind Manager* script to it.
- Right click in the *Project* window and in the context menu go to *Create* -> *Nicrom* and click on *Wind Profile*. A WP is a scriptable object that contains the global properties of the wind. If you already have an WP asset, you do not need to create a new one.
- Drag and drop the WP asset in the *Profile* property of the *Wind Manager*, Figure 3.
- Drag and drop the scene main light in the *Main Light* property.



Figure 3

The properties of a *Wind Profile* are divided in 3 tabs: *Vegetation*, *Water* and *Sky*, Figure 4.

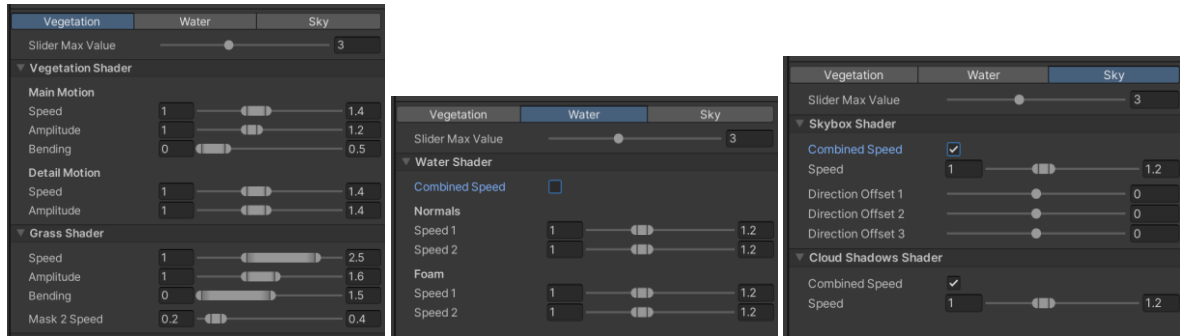


Figure 4

2.3. Post Processing Volume Manager

The *PPV Manager* controls the weights of 4 Post Processing Volumes, that are divided in 2 groups. Primary Volumes: *day*, *night* and Secondary Volumes: *sunset* and *sunrise*. The secondary volumes can be disabled if not needed. To add the *PPV Manager* to a scene:

- Create an empty game object and add *PPV Manager* script to it.
- Create 4 empty game objects and place them as children of the *PPV Manager* script.
- Add a *Post Processing Volume* to each game object.
- Assign a *Post Processing Profile* to the 4 Volumes.
- Drag and drop the 4 game objects in the appropriate fields in the *PPV Manager*, Figure 5.
- Drag and drop the *Environment Manager* game object in the *Environment Manager* Property.

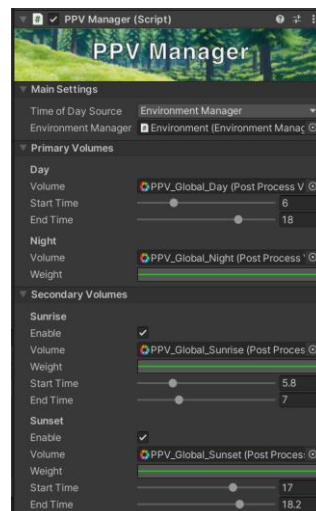


Figure 5

2.4. SGP Manager

The *SGP Manager* controls the properties of *Distance Fade* and enables/disables the *Blending of the Terrain Color* with the grass and flowers. To add the *SGP Manager* to a scene:

- Create an empty game object and add *SGP Manager* script to it, Figure 6.

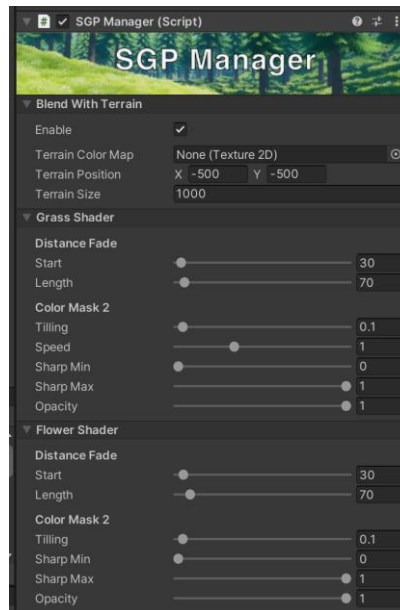


Figure 6

For *Blend with Terrain* feature to work correctly you need to specify in the *SGP Manager* properties, the size and position of the terrain, and provide a reference to the texture that contains the color information of the terrain. The terrain color texture can be generated using the [Terrain Data Baker](#) editor tool found on the Asset Store.

3. Support for third party models

By default, the shaders that are included in the assets created by Nicrom will not work with third party models. To make them work, some modifications to the models will need to be done in Blender or other 3D software.

3.1. Local Pivot Points

In order for the wind simulation to work correctly the *grass*, *flower* and *water surface plants* shaders needs to know the local pivot point positions. This information needs to be stored in the second UV channel of a model. To bake the local pivots into the second UV channel:

- Import the model in Blender.
- Add a second UV map if it doesn't have one already, Figure 7
- Enter model Edit mode. Make sure the second UV map is selected.

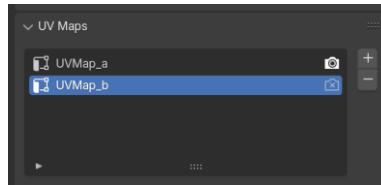


Figure 7

- In the UV Editor, press **A** key to select all the UVs. Press **S** key then **O** followed by **Enter**. This will collapse the UV islands to a single point.
- Change *Selection mode* to *Edge*. Select the edge for which you want to find the local pivot, Figure 8.

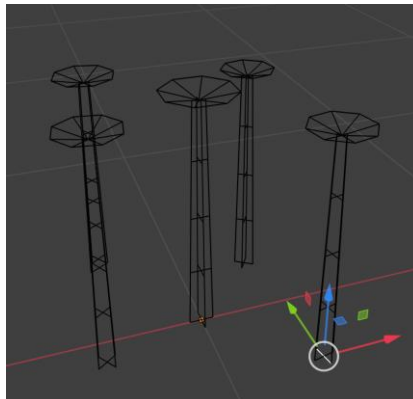


Figure 8

- Press **N** key to open the sidebar if it is hidden and select *Item* properties panel.
- In the *Transform* properties, take a look the *Median*: X and Y values, Figure 9. These are the values we need to bake into the second UV channel.

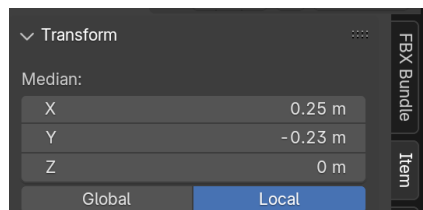


Figure 9

- Select all the vertices that will use the position in Figure 9 as their pivot point, Figure 10
- Change Blender workspace to *UV Editing*.
- Press **N** key to open the sidebar if it is hidden.
- Select *Image* properties panel, Figure 11.
- Copy *Median X* value from Figure 9 and paste it in *UV Vertex X*, Figure 11.
- Copy *Median Y* value and paste it in *UV Vertex Y*.
- Repeat the above steps for the other parts of the model.

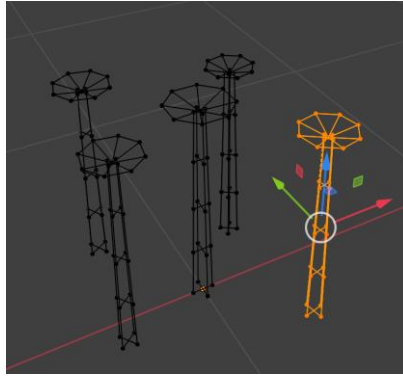


Figure 10

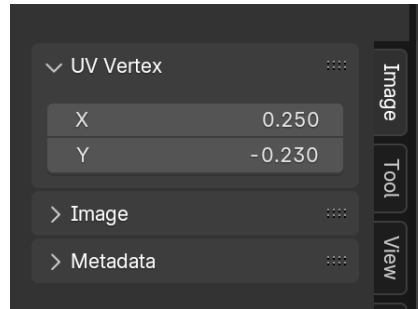


Figure 11

3.2. Branch/Leaf Rotation Angle and Pivot Point

For *Detail Motion* to work correctly the shaders need to know the pivot point position on the Y axis (Unity) of a branch (the point where the branch intersects the trunk) and its angle of rotation on the XZ axis (Unity). These 2 values will be baked into UV Channel 3. To find the rotation angle of a branch on the XY axis (Blender):

- Import the model in Blender.
- If the model has only a one UV map, add 2 new UV maps, Figure 12.

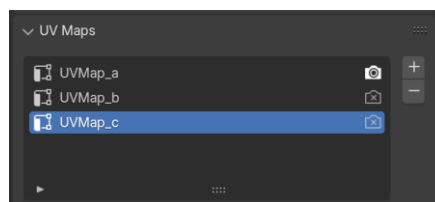


Figure 12

- Create a plane and place it at the same position as the tree model.
- Add a material to the plane.
- Assign the protractor image that is included with Nicrom assets, to the plane, Figure 13.
- Enter model Edit mode. Make sure the third UV map is selected, Figure 12.

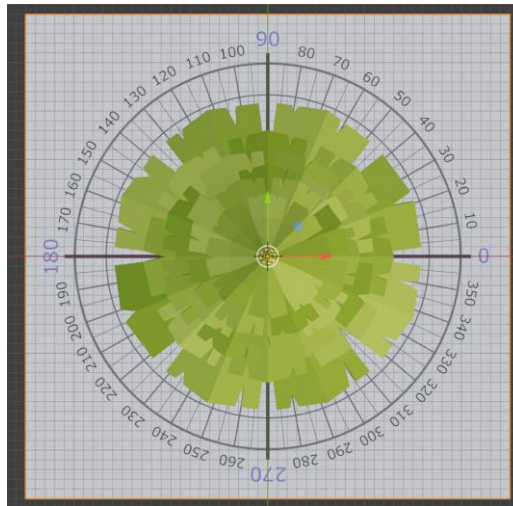


Figure 13

- Change Blender workspace to *UV Editing* and Press **N** key to open the sidebar if it is hidden.
- In the UV Editor, press **A** key to select all the UVs. Press **S** key then **O** followed by **Enter**. This will collapse the UV islands to a single point.
- Switch to Top View mode and use the protractor image to find the angle of a branch. As an example, the branch in Figure 14 has an angle of around 21 degrees.

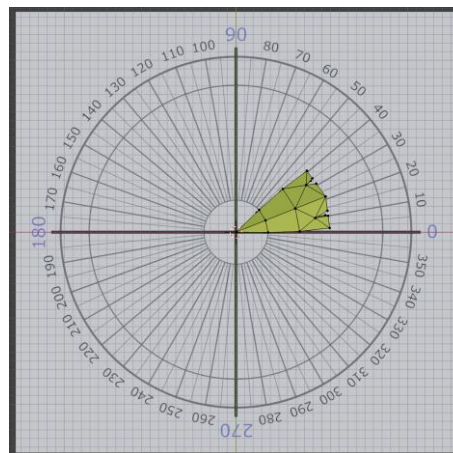


Figure 14

- Select all the vertices of the branch for which you want to bake the rotation angle.
- In *Image* panel, *UV Vertex* properties, set **X** value to the angle of the branch divided by 360. As an example, the branch in Figure 14 has a rotation angle of 21 degrees, so the value that needs to be put in X is $21/360$. This will give us a value of 0.058, Figure 15.

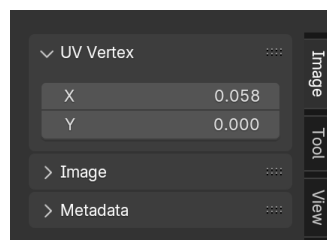


Figure 15

- Switch to front view mode and enable wireframe. The pivot point for the branch with the rotation angle of 21 degrees, looks to be at position 3.8 on the Z axis (Blender), Figure 16.

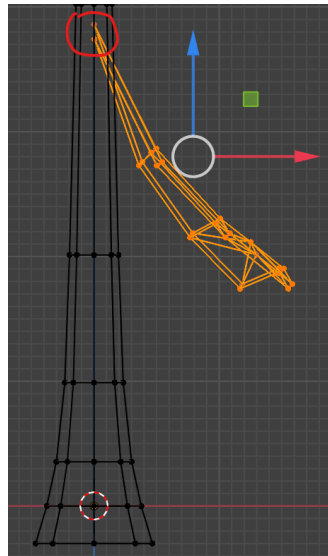


Figure 16

- In the UV Editor, set the *UV Vertex Y* value to be the same as the position of the branch pivot point, Figure 17.

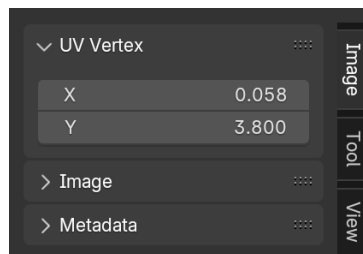


Figure 17

- Repeat the above steps for other branches of the model.

3.3. Branch/Leaf Phase Shift

The Phase Shift is stored in the *Alpha* channel of a Vertex Color of a branch and is used by the shader to add per branch motion variation.

By default, it is a little difficult to change a specific Vertex Color channel value in Blender, so we will use an add-on called [Vertex Colors Plus](#) to do this. The final result should look something like in Figure 18, where neighbouring branches have opposite alpha values.



Figure 18

To preview the alpha values of the Vertex Color us the *Color Attribute* node, Figure 19

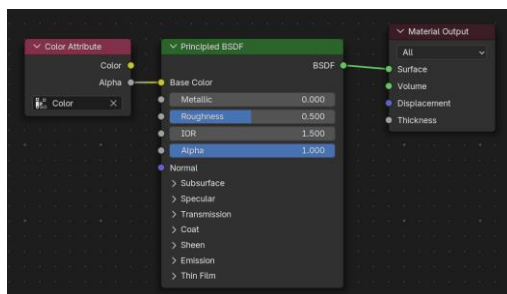


Figure 19

3.4. Detail Motion Masks

The positions of the UVs in the first UV channel are used as a detail motion mask.

- Select the vertices of all the branches.
- Change Blender workspace to *UV Editing* and Press **N** key to open the sidebar if it is hidden.
- Select all the branch UVs. Make sure the first UV Map is selected.
- Move the UVs 2 positions up to enable Detail Motion 1 and 2 positions to the right to enable Detail Motion 2, Figure 20

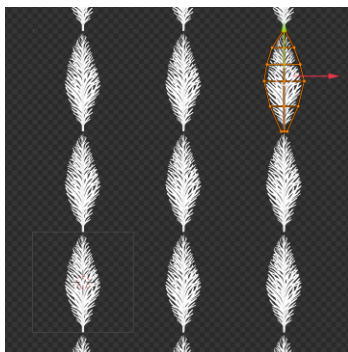


Figure 20

4. Notes

- The models that use shaders created by Nicrom must not have **Batching Static** enabled. When static batching is enabled Unity will try to merge different models into one mesh and this will generate a new pivot point.
- The grass, flowers and water surface plants must not have **Contribute GI** toggle enabled as this will override the second UV channel. The second UV channel is used to store the local pivots.

5. Social Links

- **Discord:** <https://discordapp.com/invite/RCdETwg>
- **Email:** johnq002@gmail.com
- **Facebook:** <https://www.facebook.com/Nicrom3D>